

HW3

1) $U(c_1, c_2)$

Part 1

$$1) a = y_1 - c_1$$

$$c_2 = y_2 + (1+r)a \rightarrow a = \frac{c_2 - y_2}{1+r}$$

$$y_1 - c_1 = \frac{c_2 - y_2}{1+r}$$

$$\left[y_1 + \frac{y_2}{1+r} = c_1 + \frac{c_2}{1+r} \right]$$

$$2) 1 = \log(c_1) + \beta \log(c_2) + \lambda \left(y_1 + \frac{y_2}{1+r} - c_1 - \frac{c_2}{1+r} \right)$$

$$[c_1]: \frac{1}{c_1} - \lambda = 0 \rightarrow \frac{1}{c_1} = \lambda$$

$$[c_2]: \frac{\beta}{c_2} - \frac{\lambda}{1+r} = 0 \rightarrow \frac{\beta(1+r)}{c_2} = \lambda \rightarrow \frac{1}{c_1} = \frac{\beta(1+r)}{c_2} \rightarrow \boxed{\frac{c_2}{c_1} = \beta(1+r)}$$

$$[\lambda]: y_1 + \frac{y_2}{1+r} = c_1 + \frac{c_2}{1+r}$$

$$3) c_2 = c_1 \beta(1+r)$$

$$y_1 + \frac{y_2}{1+r} = c_1 + \frac{c_1 \beta(1+r)}{1+r}$$

$$1) \frac{y_1 + \frac{y_2}{1+r}}{1+\beta} = c_1 \rightarrow \frac{200 + \frac{105}{1.05}}{1.95} = \boxed{153.846}$$

$$\left(\frac{y_1 + \frac{y_2}{1+r}}{1+\beta} \right) \beta(1+r) = c_1 \beta(1+r)$$

$$2) \frac{\beta(1+r) y_1 + \beta y_2}{1+\beta} = c_2 \rightarrow 153.846 \cdot (.95) \cdot (1.05) = \boxed{153.462}$$

$$3) a = y_1 - c_1 \rightarrow 200 - 153.846 = \boxed{46.153}$$

Part 2:

$$1) S = (1+r)T$$

$$a = y_1 - T - c_1$$

$$c_2 = y_2 + S + (1+r)a \rightarrow a = \frac{c_2 - y_2 - S}{(1+r)} \rightarrow a = \frac{c_2 - y_2 - (1+r)T}{1+r}$$

$$y_1 - T - c_1 = \frac{c_2 - y_2}{1+r} - T$$

$$y_1 + \frac{y_2}{1+r} = c_1 + \frac{c_2}{1+r}$$

Tax Doesn't Affect The Intertemporal Budget Constraint

2) Given $y_1 = 200$ $y_2 = 105$ $B = .95$ $r = .05$ and $T = 20$

c_1 and c_2 unchanged because same utility and constraint

$$a = y_1 - T - c_1 = 200 - 20 - 153.846 = \boxed{26.154}$$

The Lump Sum Tax doesn't impact c_1 or c_2 but does impact saving

Part 3: $S(c_1) = (1+r) T c_1$

1) $a = y_1 - c_1(1+r)$

$$c_2 = y_2 + S(c_1) + (1+r)a \rightarrow a = \frac{c_2 - y_2 - S(c_1)}{1+r}$$

$$\frac{c_2 - y_2 - S(c_1)}{1+r} = y_1 - c_1(1+r)$$

$$y_1 + \frac{y_2}{1+r} = c_1(1+r) + \frac{c_2}{1+r} - \frac{S(c_1)}{1+r}$$

$$y_1 + \frac{y_2}{1+r} = c_1(1+r) + \frac{c_2}{1+r} - \frac{(1+r) T c_1}{1+r}$$

$$\boxed{y_1 + \frac{y_2}{1+r} = c_1 + \frac{c_2}{1+r}} \quad \text{No change in Intertemporal Budget Constraint}$$

2) c_1, c_2 same as in Part 1. Part 2, Given $\tau = .05$

$$a = y_1 - c_1(1.05)$$

$$= 200 - 153.846(1.05) = \boxed{38.4617}$$

$$1 - s + \frac{s}{1+r}$$

$$1+r - s(1+r) + s$$

$$\frac{1+r-sr}{1+r}$$

$$\text{Part 4: } S(c_1) = (1+r)T$$

$$S(c_1) = s \cdot c_1$$

$$a = y_1 - c_1 - T$$

$$c_2 = y_2 + s c_1 + (1+r)a$$

$$S \cdot c_1 = \frac{(1+r)T}{1+r} = \frac{s c_1}{1+r}$$

$$1) \dots y_1 - c_1 - T = \frac{c_2 - y_2 - s c_1}{1+r}$$

$$y_1 + \frac{y_2}{1+r} = c_1 + \frac{c_2}{1+r} - \frac{s c_1}{1+r} + T$$

$$\mathcal{L} = \log(c_1) + \beta \log(c_2) + \gamma \left(y_1 + \frac{y_2}{1+r} - c_1 - \frac{c_2}{1+r} + \frac{s c_1}{1+r} - T \right)$$

$$c_1] \frac{1}{c_1} + \gamma \left(-1 + \frac{s}{1+r} \right) = 0 \Rightarrow \frac{1+r}{c_1} = \gamma (1+r-s)$$

$$c_2] \frac{\beta}{c_2} = \frac{\gamma}{1+r} \Rightarrow \left(\frac{c_1}{c_2} \right)^\beta = \gamma \Rightarrow \frac{(1+r)\beta}{c_2} = \frac{1+r}{c_1(1+r-s)}$$

$$\frac{c_2}{c_1} = \beta(1+r-s)$$

$$\text{Question 2: } U(c, l) = \log c + \gamma \log(1-l)$$

$$c = wl + t$$

$$1) \mathcal{L} = \log c + \gamma \log(1-l) + \gamma(wl + t - c)$$

$$2) \frac{1}{c} = \gamma$$

$$1] - \frac{\gamma}{1-l} = -w\gamma \rightarrow \frac{1}{1-l} = \frac{\gamma}{(1-l)w} \rightarrow c = \frac{(1-l)w}{\gamma}$$

$$\gamma] wl + t = c$$

$$wl + t = \frac{(1-l)w}{\gamma}$$

$$\gamma l + \frac{\gamma t}{w} = 1-l$$

$$\frac{\gamma t}{w} - 1 = -l - \gamma l$$

$$\frac{1 - \frac{\gamma t}{w}}{1+\gamma} = l$$

$$\frac{w - \gamma t}{w(1+\gamma)}$$

$$c = \left(1 - \frac{1 - \frac{\gamma t}{w}}{1+\gamma} \right) w = \frac{w - \frac{\gamma t}{1+\gamma}}{\gamma} = \frac{\gamma w - \gamma t}{\gamma(1+\gamma)} = \frac{w - t}{1+\gamma}$$

$$-t w - t w^2 - w^2 + t w^2$$

$$-(t+w)w(1+\gamma)$$

$$\frac{-t(w(1+\gamma)) - w(w - \gamma t)}{w^2(1+\gamma)^2}$$

3) If t increases, l decreases

4) If γ increases, l decreases ($\frac{\partial l}{\partial \gamma}$ is negative)

$$5) \max AL^D - wL^D$$

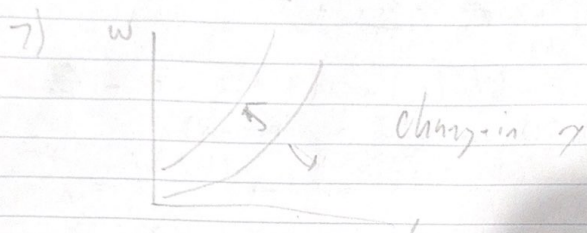
$$\text{st } y = AL^D$$

$$10) A - w = 0$$

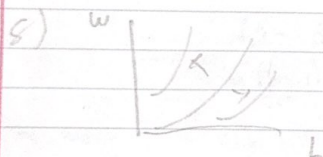
$$w = A$$

$$6) L^S = \frac{A - \gamma E}{A(1 + \gamma)}$$

(assuming problem means labor instead of wage)
Wage = A



$\gamma \uparrow$ then $L \downarrow$
 $\gamma \downarrow$ then $L \uparrow$



$E \uparrow$ then $L \downarrow$
 $E \downarrow$ then $L \uparrow$

9) $\frac{dL}{dA} :$ $\frac{A(1 + \gamma) - (A - \gamma E)(1 + \gamma)}{A^2(1 + \gamma)^2}$

$A \uparrow$ then $L \uparrow$
 $A \downarrow$ then $L \downarrow$

$$\frac{\gamma E}{A^2(1 + \gamma)} < 0$$